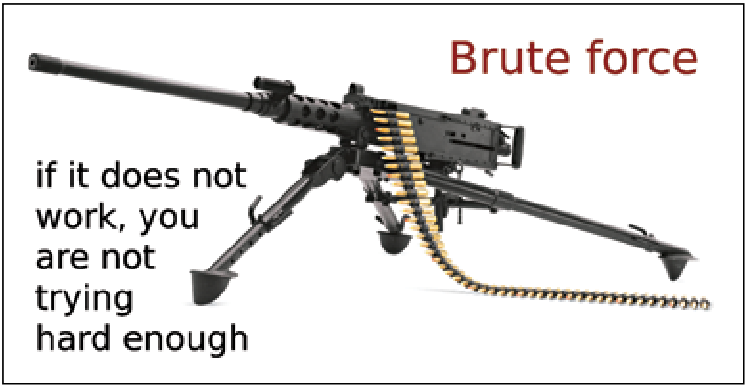

A brute-force attack is guaranteed to break a cipher eventually

Under normal circumstances, breaking a public key cipher via brute force might take a lot of time. But cyber warfare is rarely a normal situation. If an attack is conducted by a very large organization or a country, it is possible that they have enough computing infrastructure that they can attempt a brute force on medium-level security. In case of nation sponsored attacks, it is possible to include mathematical research and expertise to aid the brute force.

Role of Computing hardware in breaking ciphers: If you think the processor in a normal PC can break a cipher using brute force that governments might be using, then you are not wrong, not at least technically. Normal processors available in market can be used to break ciphers quite well however, they are not good enough to do it within limited (read useful) time period. This is because the CPU is designed to perform parallel processing only to a certain degree.

GPGPU technologies such as CUDA, OpenCL and DirectCompute can be used to accelerate the process by thousands of times. If one is to take the help of advanced databases and previous research done in the field, the process can run upto a million times faster. If you dive into the architectures of some of the top supercomputers of this day, you will find GPUs in use. The current fastest supercomputer Titan uses GPUs from both the red and green groups (we mean AMD/ATi and Nvidia). That does not mean that it is all insecure. Strength of a cipher usually increases exponentially with linear increase in strength of keys. Most communication is done using strong enough keys and it is still difficult to fetch your Gmail password